

# RandomForest — Model Evaluation

| field                          | value                  |
|--------------------------------|------------------------|
| model_label                    | RandomForest           |
| estimator_class                | RandomForestClassifier |
| param.bootstrap                | True                   |
| param.ccp_alpha                | 0.0                    |
| param.class_weight             | None                   |
| param.criterion                | gini                   |
| param.max_depth                | 10                     |
| param.max_features             | sqrt                   |
| param.max_leaf_nodes           | None                   |
| param.max_samples              | None                   |
| param.min_impurity_decrease    | 0.0                    |
| param.min_samples_leaf         | 1                      |
| param.min_samples_split        | 2                      |
| param.min_weight_fraction_leaf | 0.0                    |
| param.monotonic_cst            | None                   |
| param.n_estimators             | 300                    |
| param.n_jobs                   | None                   |
| param.oob_score                | False                  |
| param.random_state             | 0                      |
| param.verbose                  | 0                      |
| param.warm_start               | False                  |

RandomForest — SUMMARY OUTPUT (Classification)

| Metric            | Value                    |
|-------------------|--------------------------|
| Test observations | 2348                     |
| Accuracy          | 0.5315                   |
| Balanced accuracy | 0.4687                   |
| Macro precision   | 0.4765                   |
| Macro recall      | 0.4687                   |
| Macro F1          | 0.4163                   |
| Weighted F1       | 0.4688                   |
| Dominant error    | 0 (Loss) → 2 (Win) [281] |

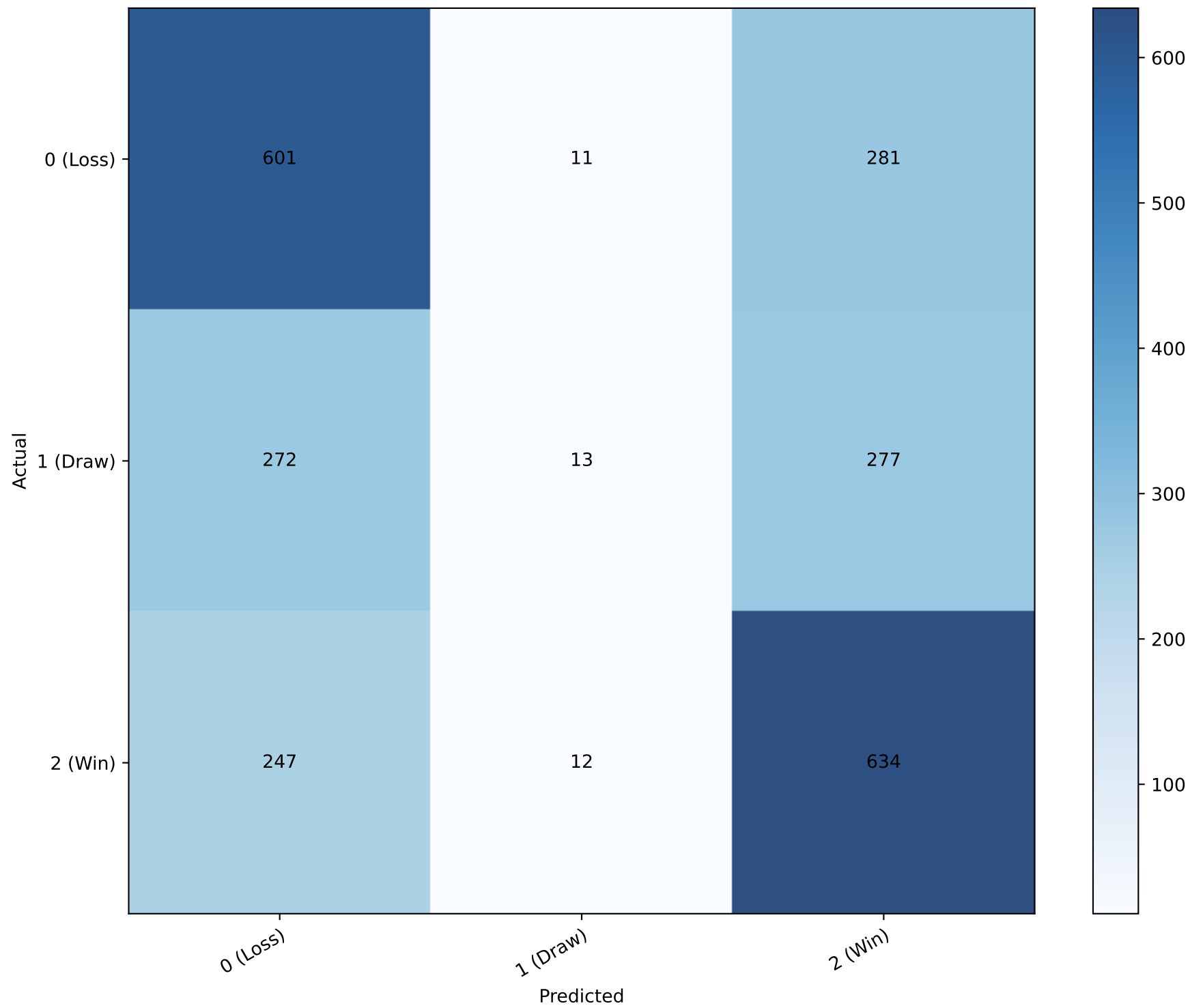
RandomForest — Metrics Summary

| metric             | value               |
|--------------------|---------------------|
| accuracy           | 0.5315161839863713  |
| balanced_accuracy  | 0.46870346533404    |
| precision_macro    | 0.4765324828663755  |
| recall_macro       | 0.46870346533404    |
| f1_macro           | 0.4162501554513533  |
| precision_weighted | 0.49280355358655975 |
| recall_weighted    | 0.5315161839863713  |
| f1_weighted        | 0.46880019680849633 |
| n_test             | 2348.0              |

RandomForest — Classification Report

| class_or_avg | precision           | recall               | f1-score             | support            |
|--------------|---------------------|----------------------|----------------------|--------------------|
| 0            | 0.5366071428571428  | 0.6730123180291153   | 0.5971187282662692   | 893.0              |
| 1            | 0.3611111111111111  | 0.023131672597864767 | 0.043478260869565216 | 562.0              |
| 2            | 0.5318791946308725  | 0.70996640537514     | 0.6081534772182254   | 893.0              |
| accuracy     | 0.5315161839863713  | 0.5315161839863713   | 0.5315161839863713   | 0.5315161839863713 |
| macro avg    | 0.4765324828663755  | 0.46870346533404     | 0.4162501554513533   | 2348.0             |
| weighted avg | 0.49280355358655975 | 0.5315161839863713   | 0.46880019680849633  | 2348.0             |

RandomForest — Confusion Matrix



## RandomForest — Feature Importances (Top 13)

| feature            | importance           |
|--------------------|----------------------|
| b365_team_win_prob | 0.42009780408640224  |
| b365_draw_prob     | 0.07233903687681839  |
| b365_home_prob     | 0.05970421847413986  |
| b365_away_prob     | 0.05724659084340466  |
| xGA_last3          | 0.05312932158653426  |
| xGA_last5          | 0.05296318531917114  |
| xG_last3           | 0.052048777901617195 |
| xG_last5           | 0.0507119722113329   |
| pts_last5          | 0.040261837625007735 |
| B365H              | 0.03797478514030609  |
| B365D              | 0.03564206057185844  |
| B365A              | 0.03491502602806135  |
| pts_last3          | 0.032965383335345655 |